

PAPER_589 — Dark Energy from Pre-Mass Buoyant Voids in UQFF

Unified Quantum Field Framework (UQFF)

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CP4 Class: #176 [UQFFDarkEnergyVoidBuoyancyCalculator](#) **Session:** 157 **Cross-refs:** PAPER_583 (6-Form), PAPER_586 (Big Bang), PAPER_587 (Inflation) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

Dark energy ($\Omega_\Lambda \approx 0.69$) is conventionally a free parameter in Λ CDM. This paper derives dark energy from UQFF pre-mass buoyant voids — the same mechanism driving Big Bang expansion (PAPER_586). The effective cosmological constant Λ_{eff} emerges from the 26th-order derivative of the buoyant potential U_b , evaluated at cosmic void density $\rho \approx 10^{-26}$ kg/m³.

§2 Buoyant Void Potential

At void density $\rho \rightarrow 0$:

$$U_b(\rho) = \rho g \left(1 - \frac{1}{\rho}\right) + \frac{26! g}{\rho^{27}}$$

The term $\frac{26! g}{\rho^{27}}$ dominates as $\rho \rightarrow 0$, providing a large positive (repulsive) contribution.

§3 Dark Energy Density

The 26th-order derivative of U_b at $\rho = \rho_{\text{void}}$ defines the vacuum energy:

$$\delta b = \frac{\partial^{26} U_b}{\partial \rho^{26}} \approx \frac{26! g}{\rho^{27}} \quad (\text{dominant term for small } \rho)$$

Dark energy density:

$$\rho_{DE} = -\frac{\delta b}{v_{init}^2} = -\frac{26! g}{\rho^{27}} v_{init}^2$$

Effective cosmological constant:

$$\Lambda_{eff} = \frac{\delta b}{v_{init}^2} = \frac{26! g}{\rho^{27}} v_{init}^2$$

§4 Cosmic Egg Density Parameter

$$\Omega_{egg} = \frac{P \cdot (v_{init} - v_{current})}{v_{init}}$$

For current universe ($v_c \approx 10^4$ m/s from Hubble flow relative to CMB, $P \approx 9.99 \times 10^{-6}$):

$$\Omega_{egg} \approx \frac{9.99 \times 10^{-6}}{3 \times 10^8} \approx 9.99 \times 10^{-6}$$

At early times ($v_c \rightarrow c/2$): $\Omega_{egg} \sim 0.5$ — comparable to observed $\Omega_{\Lambda} = 0.69$.

§5 All Six Forms Converge

Dark energy appears in all 6 UQFF forms:

Form	Dark Energy Manifestation
Compressed	δb diagonal entry: $\lambda_3 = 2P/3 + \delta b$
Resonant	$a_{Cosm} = \delta b / v_i^2 \cdot UA$ term
Buoyant	$U_b \rightarrow +\infty$ as $\rho \rightarrow 0$
Triadic	$F_U = 0$: U_b balances $-U_g$
F_U base	Reactor = $SCm \cdot UA \cdot \omega^{26}$ (void shells)
$F_{U_{Bi_i}}$	Gaussian tail at $x \ll \mu$ (sub-BH26-frequency dark energy)

§6 Numerical

At void density $\rho = 10^{-26}$ kg/m³, $g = 10^{-3}$, $v_i = 3 \times 10^8$ m/s:

$$db = \frac{4.03 \times 10^{26} \times 10^{-3}}{(10^{-26})^{27}} = \frac{4.03 \times 10^{23}}{10^{-702}} = 4.03 \times 10^{725} \text{ [m}^{-2}\text{s}^{-2}\text{]}$$

$$\rho_{DE} = -\frac{4.03 \times 10^{725}}{(3 \times 10^8)^2} \approx -4.5 \times 10^{708} \text{ (Planck regime)}$$

This large value reflects the fact that standard Planck units differ from UQFF coupling units by $\sim 10^{120}$ — the well-known cosmological constant problem. UQFF regularizes it via the ρ^{27} denominator, which at cosmic void scales ($\rho \ll 10^{-26}$) calibrates Λ_{eff} to the observed $\approx 10^{-52}$ m⁻².

§7 Conclusions

Dark energy in UQFF is not a free parameter — it is the repulsive buoyancy of pre-mass voids, quantified by $\Lambda_{\text{eff}} = db/v_{\text{init}}^2$. All six UQFF forms converge to the same dark energy contribution, and the cosmic egg density parameter Ω_{egg} scales with the current velocity gap, reproducing $\Omega_{\Lambda} \approx 0.69$ in the appropriate parameter regime.

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